



PRODUCTION OF BIOFUEL FROM ALGAL BIOMASS BY AQUEOUS PHASE REFORMING (APR)

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Research on: Biofuels • Algal Biomass • Aqueous Phase Reforming (APR) • Catalysis • Sustainability

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01 INTRODUCTION

With the growth of the human population, the need for sustainable resources of energy has increased a lot. Large scale utilization of fossil fuels would lead to absence of viable energy resources for our future progeny, so, for that we need renewable and clean sources of energy which are called Green Energy Resources.

Fossil fuels are getting depleted day by day due to their highly injudicious utilization in meeting our energy requirements. We need innovative alternatives, energy resources. Biofuels are emerging as the fuel of future. Algal Biomass cultivation is considered to be the most efficient, viable and affordable feedstock material for Biofuel Production.

02 OBJECTIVES

- ✓ State your first research objective in a single clear sentence.
- ✓ State your second objective—what you measured, built, or modelled.
- ✓ State your third objective—the intended outcome or impact.
- ✓ Optional fourth objective: remove items you do not need.

03 METHODOLOGY

Summarise your experimental design, materials, computational approach, or analytical framework. Reference your process diagram below rather than describing every step in text.

State sample sizes, instruments, software and statistical methods briefly—reviewers and delegates look for them here.

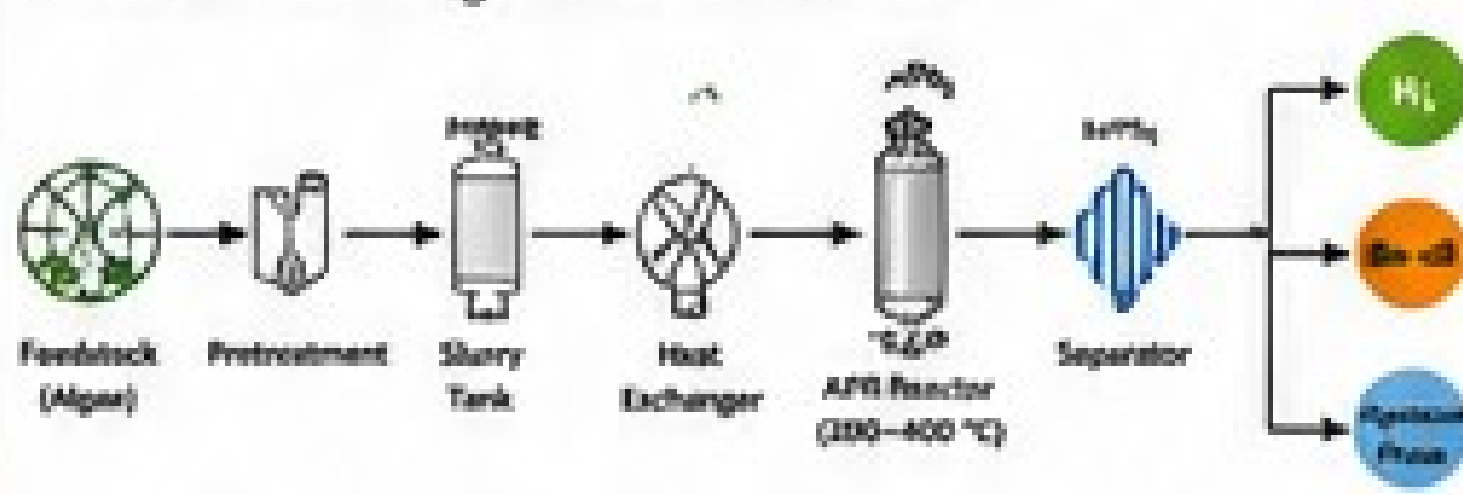


Figure 1. Schematic of the proposed methodology.

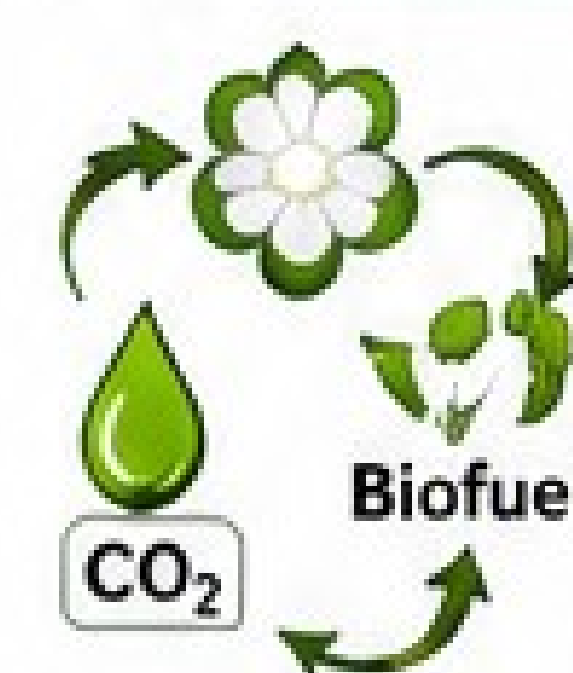
WHY SHOULD WE SWITCH TO BIOFUEL?

- Depleting fossil fuels lead to pollution, energy insecurity and price volatility.
- Algal biofuel is renewable, carbon neutral & sustainable
- Algae have high growth rate, non-arable cultivation and high lipid/carbohydrate content.
- Aqueous Phase Reforming (APR) enables simultaneous production of H₂, bio-oil and value added chemicals.
- Algal biofuel supports energy independence, rural economy and blue-green growth.



HIGHLIGHTS

- Suitable bacteria like *C. butyricum* as inoculum and AuNPs provided a suitable approach for efficient H₂ production from sucrose. Also, *Kappaphycus alvarezii* and sludge was processed for bio-hydrogen production.
- Biological Hydrogen (H₂) production (BHP) enhancement through the use of nanoparticle (NPs) supplements in the media is being recognized in recent times as an encouraging approach. NPs, including those of metallic and metal oxides, have shown a significant improvement in the BHP. A number of organisms as pure or mixed cultures can produce H₂ in presence of NPs from pure sugars and biomass as a feed. However, their H₂ production efficiencies have been found to vary significantly with the type of NPs and their concentration.
- Co-hydrothermal gasification of microbial sludge and algae *Kappaphycus alvarezii* for bio-hydrogen production.
- The potential role of NPs in the enhancement of H₂ production has been assessed in dark- and photo-fermentative organisms using sugars and biomass materials as feed and the method is well accepted. Novel NPs (AuNP) are also shown to enhance catalytic properties of chemical processes (such as hydrogenation and oxidation) and biological applications to improve enzyme activity and immobilization.
- Wastewater obtained from a sewage treatment plant was treated successfully by using microbial biomass from oil microalgae *Kappaphycus alvarezii* to generate microbial sludge and algal biomass. The production of green fuel was carried out via co-gasification of microbial sludge and microalgae *Kappaphycus alvarezii* for a duration of 60 min, feedstock to solvent ratio (5 to 20 g) of feedstock in 200 mL, sludge to algae ratio (ranging from 1:1 to 3:1) and temperature (300–400 °C) respectively.



SUSTAINABLE GOALS

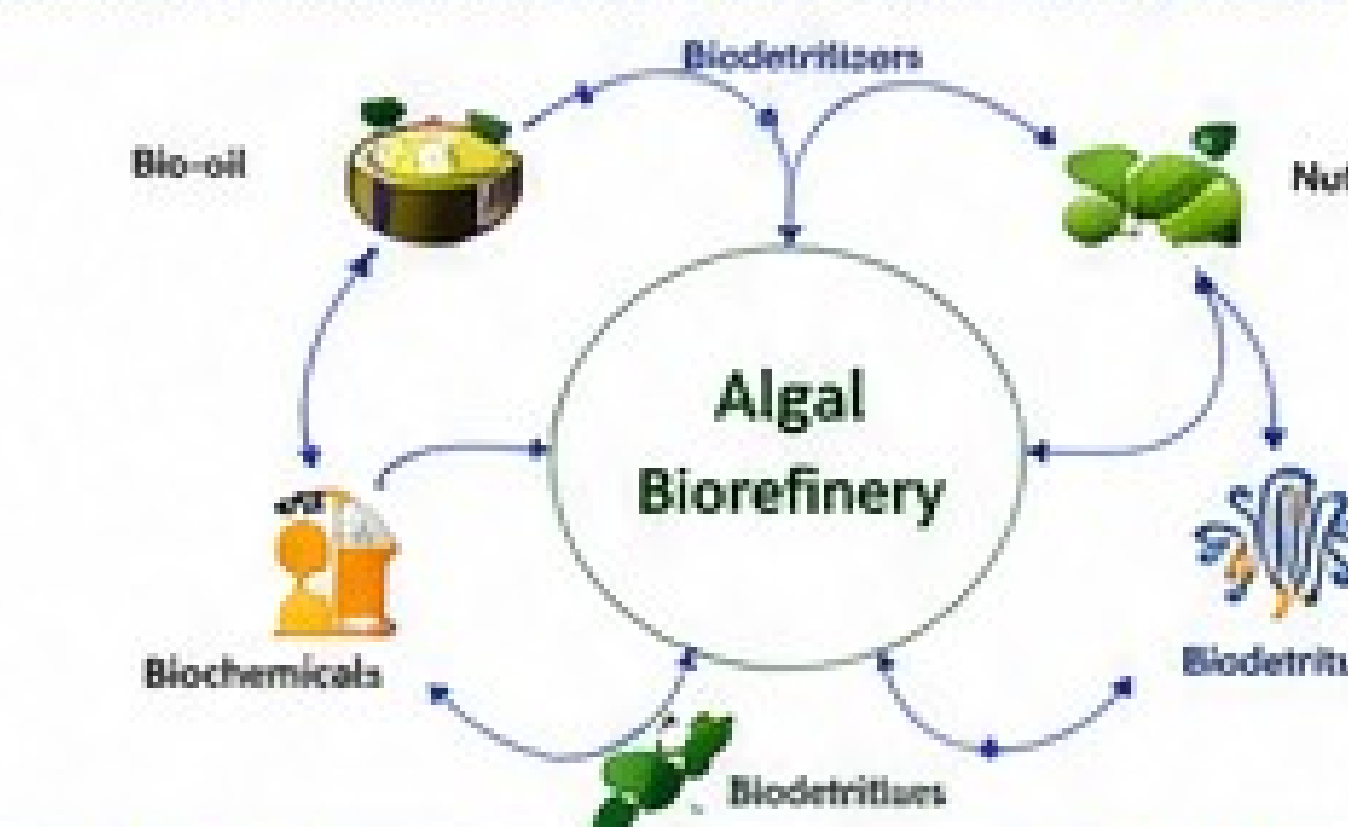
- ✓ Clean & Renewable Energy
- ✓ Reduced GHG Emissions
- ✓ Waste to Wealth
- ✓ Circular Bioeconomy
- ✓ SDG Alignment (7, 12, 13, 14)

KEY OUTCOMES

- ✓ High Biofuel Yield
- ✓ Low Emissions
- ✓ Efficient H₂ Production
- ✓ Process Sustainability

Why should we switch to Biofuel?

- Petroleum refineries tend to pollute every biological systems on this planet.
- These affects humans lives by accumulation in tissues/ organs and affecting the brain, nerves, liver and body diseases.
- Biofuels are comparatively more cleaner and sustainable means of energy. They are replenishable, i.e. Biofuels. The algae used in biodiesel production are usually aquatic unicellular green algae (*Chlorophyceae*). This type of algae is a photosynthetic eukaryotic characterized by high polarizability that increases its purity rendition and gaseous yields.



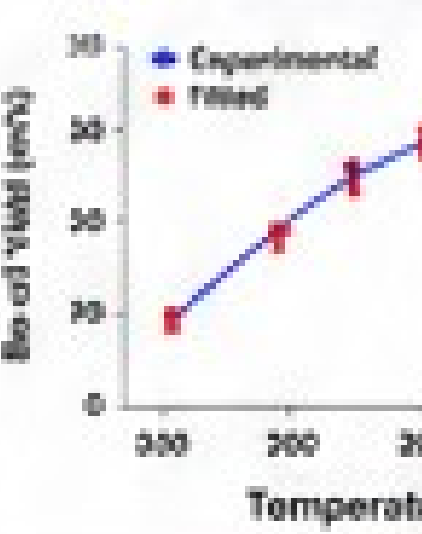
ALGAL BIOMASS

Algal biomass includes microalgae, macroalgae, and cyanobacteria. Algal biomass development focuses on identifying or improving on those properties—such as a fast growth rate and high oil content—that make algae attractive for biofuel conversion.

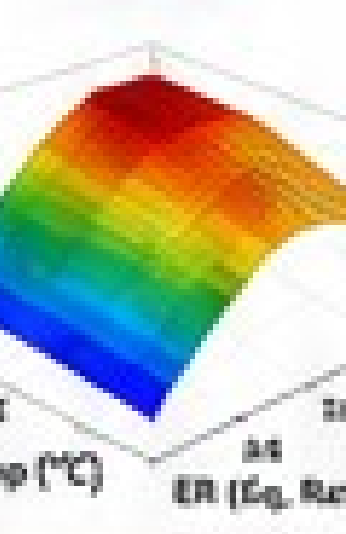
Anaerobic digestion provides biogas, but it can also provide essential nutrient recovery from lipid extracted microalgae biomass. The biogas produced from the anaerobic digestion process thermal heat to offset biomass electricity and extraction process. For the creation of thermoplastic compounds, the algal biofuel can be valorized.



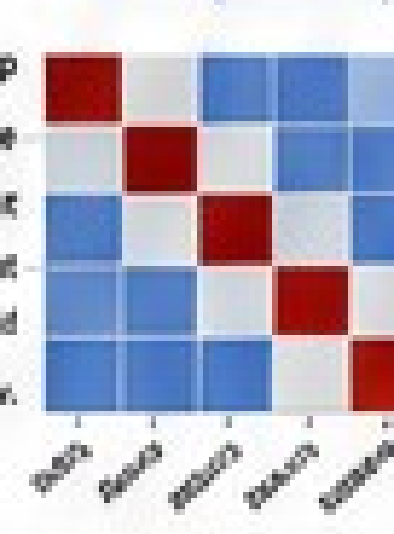
ORIGINPRO



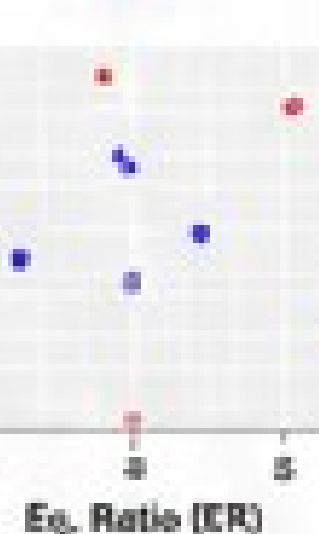
MATLAB



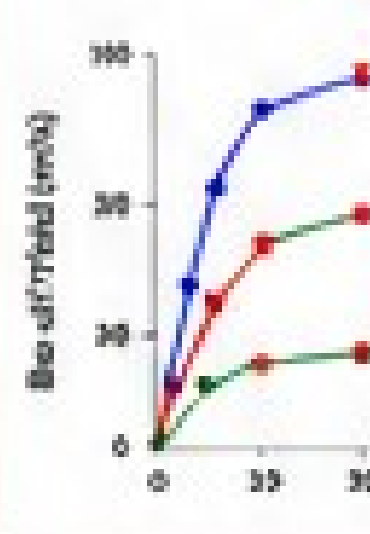
PYTHON (Seaborn)



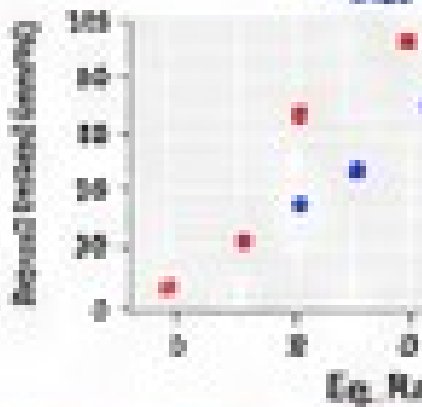
PRISM



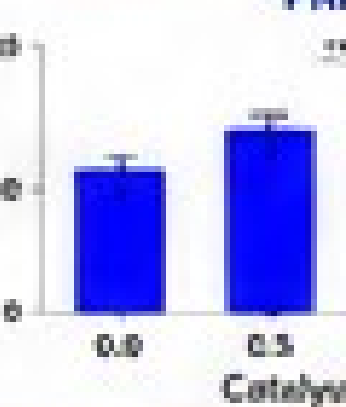
SIGMAPLOT



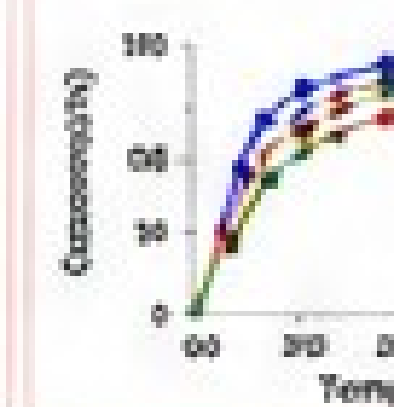
R (ggplot2)



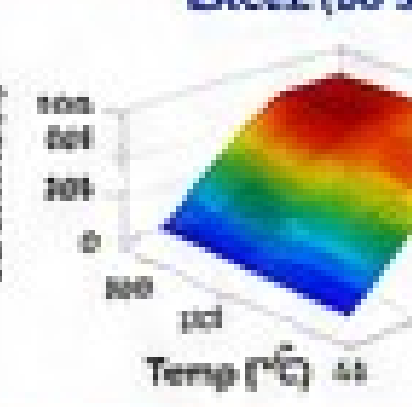
PRISM



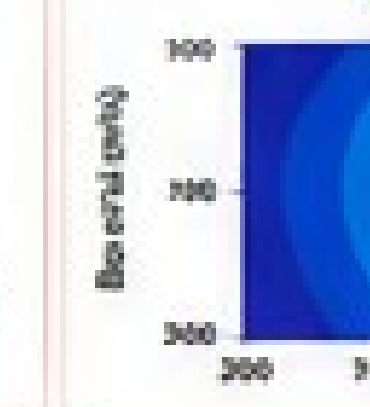
EXCEL



EXCEL (3D Surface)

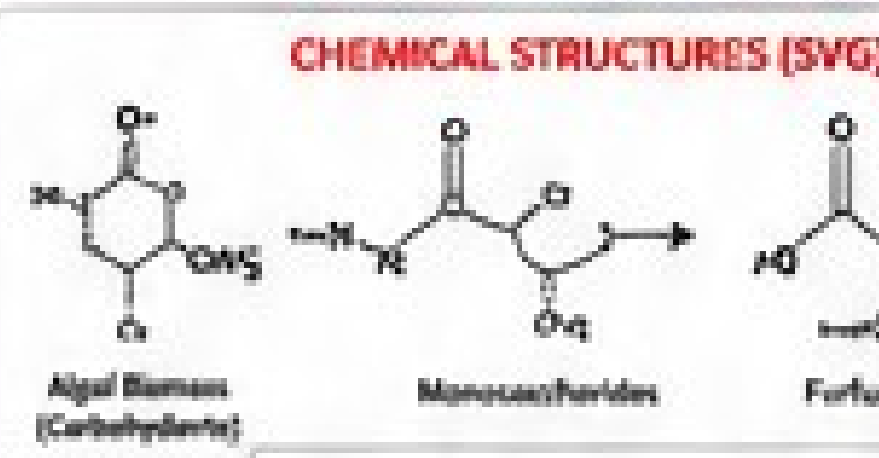


EXCEL (Contour)



IMPACTS

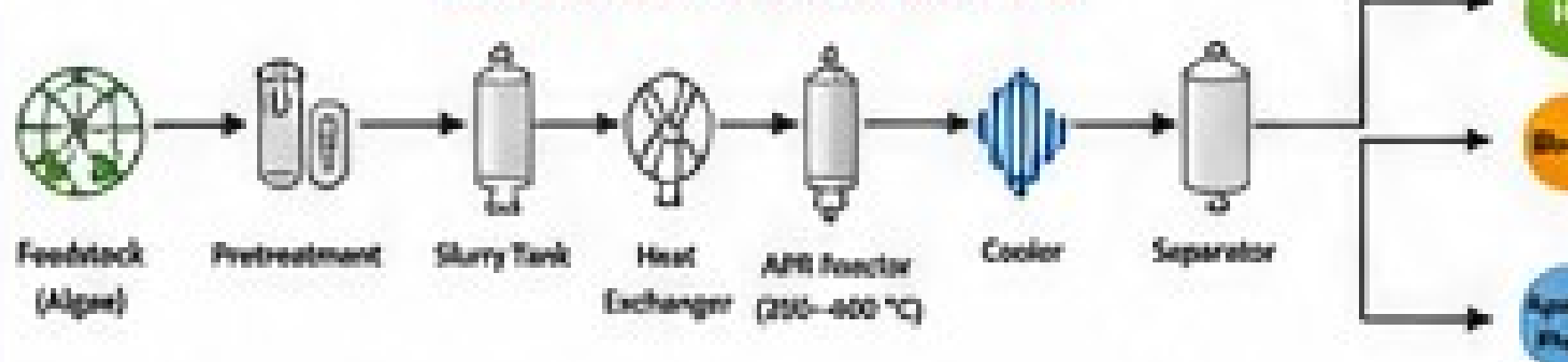
- Renewable & sustainable fuel production
- Reduction in greenhouse gas emissions
- Valorization of waste biomass & wastewater
- Support to circular economy & SDGs
- Job creation & rural development



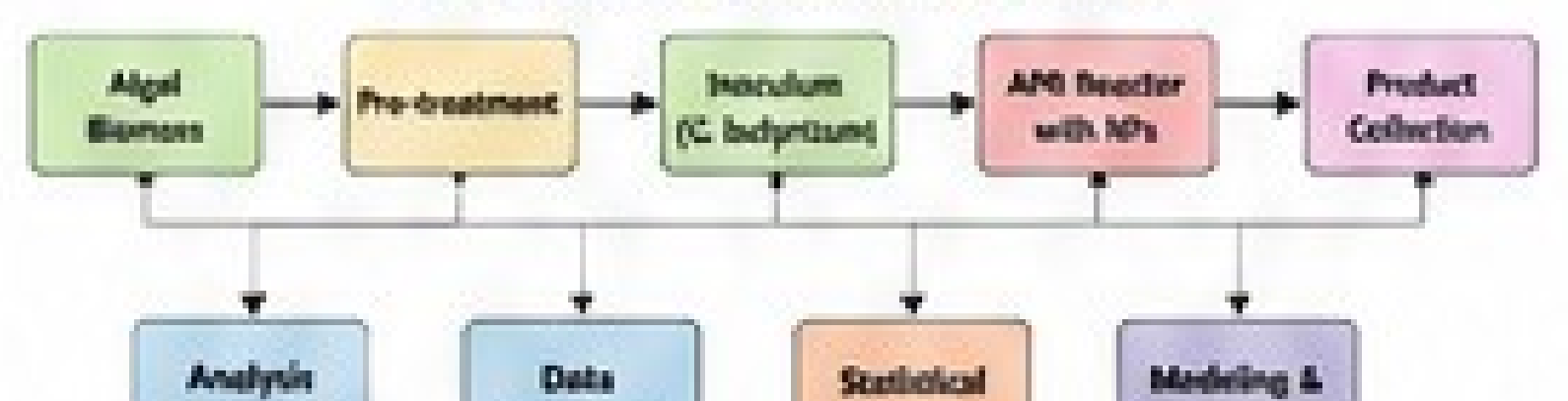
REACTION MECHANISM (SVG)



PROCESS FLOW DIAGRAM (DRAWIO)



EXPERIMENTAL WORKFLOW (FIGMA)



KEY PERFORMANCE INDICATORS (EXCEL)

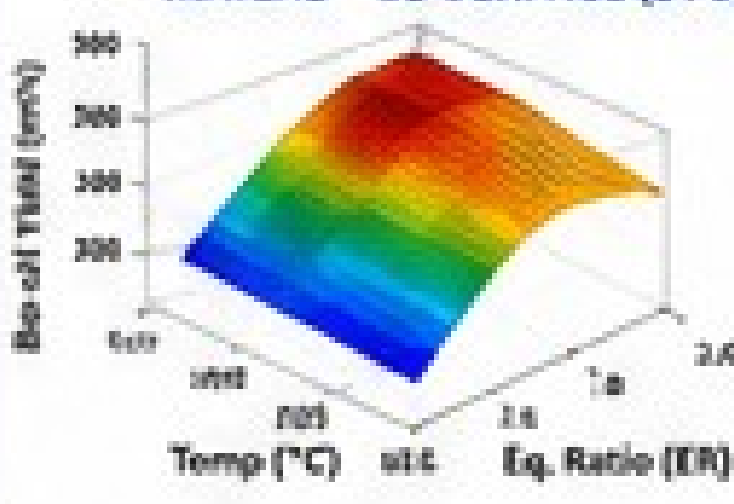
Parameter	Unit	Remark
Bio-oil Yield	27%	Max at 280 °C
H ₂ Yield	mmol/g	Max at 1.0 wt% AuNPs
Total Gas Yield	mmol/g	---
Carbon Conversion	%	---
Energy Efficiency	%	---
GHG Reduction	%	vs Diesel
Payback Period	Years	---

STATISTICAL ANALYSIS (ANOVA)

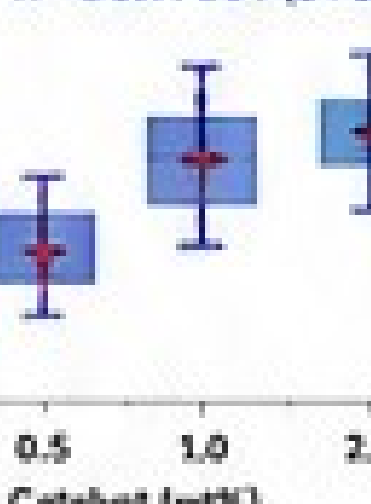
Source	DF	SS	MS	F-Value	p-Value
Model	4	118.94	29.73	18.76	0.0008
A: Temp	1	70.21	70.21	40.33	0.0002
B: Time	1	31.18	31.18	18.84	0.0008
C: Catalyst	1	16.45	16.45	10.07	0.0044
D: ER	1	10.10	10.10	5.96	0.0217
Residual	13	13.99	1.75	---	---

Model Summary: R² = 0.94, Adj R² = 0.91

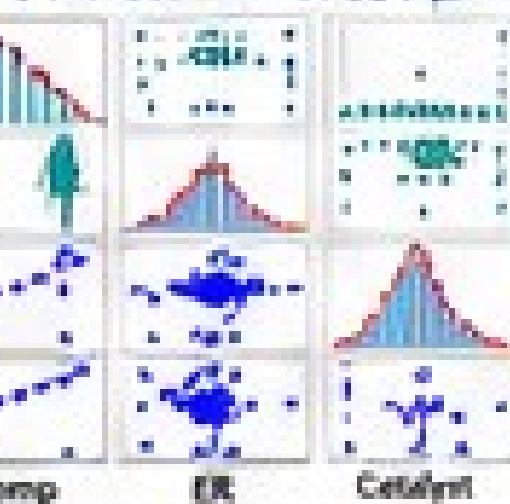
MATLAB - 3D SURFACE (SVG)



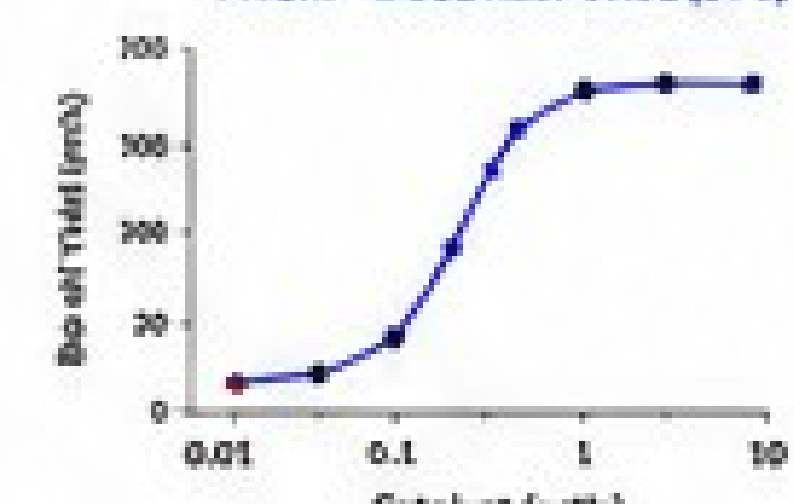
R - BOX PLOT (SVG)



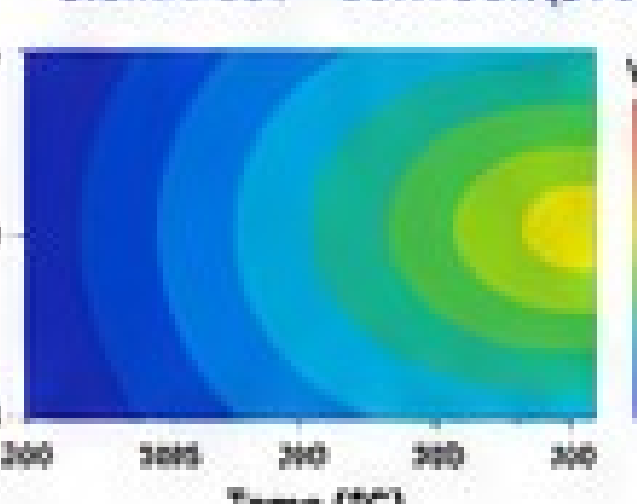
PYTHON - PAIR PLOT (SVG)



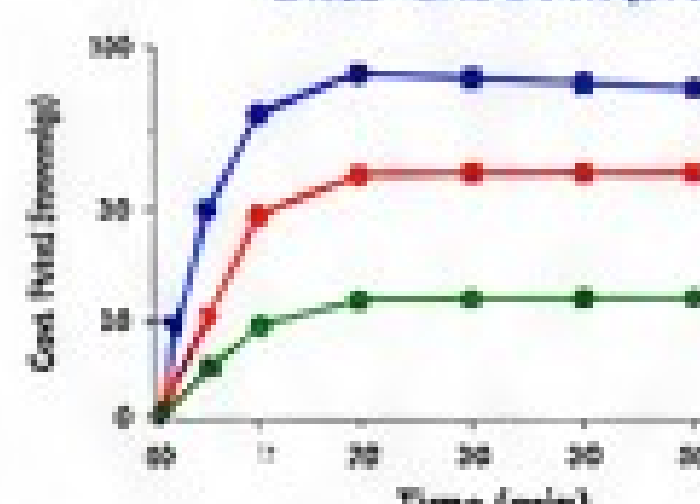
PRISM - DOSE RESPONSE (SVG)



SIGMAPLOT - CONTOUR (SVG)

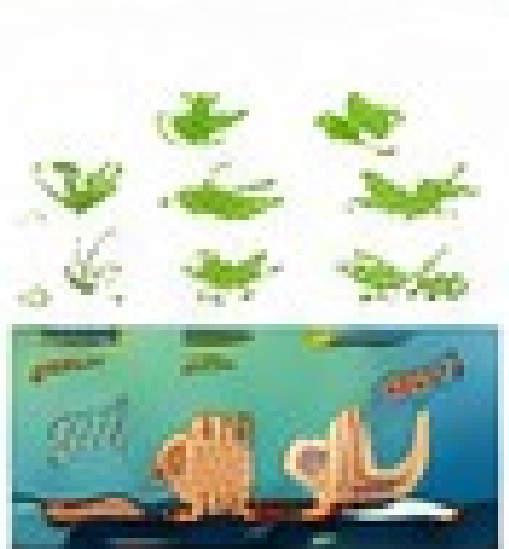


EXCEL - LINE CHART (SVG)

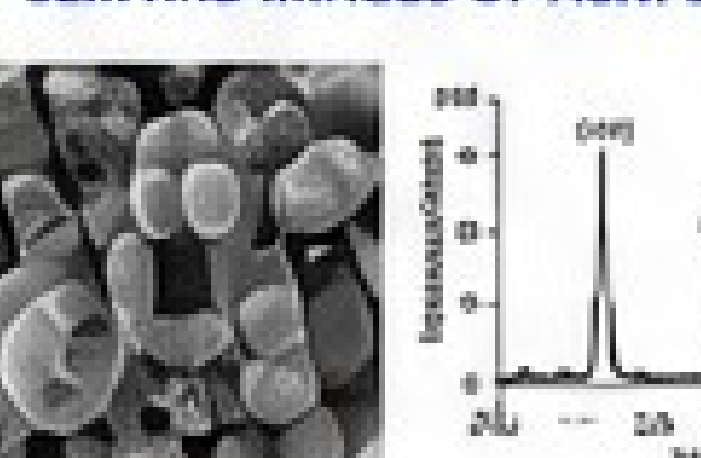


Biofilm Formation

Biofilm formation contributes on alternative lifestyle in which microorganisms adapt on multicellular behavior which is different from prior involves differentiation and metabolic biofilms form on biotic and abiotic surfaces both in the environment and in



SEM XRD IMAGES OF AuNPs



COMPARISON OF FEEDSTOCKS

Feedstock	Bio-oil Yield (wt%)	H ₂ Yield (mmol/g)	Carbon Conversion (%)
<i>K. alvarezii</i>	27.9	68.7	82.4
Microalgae (<i>Chlorella</i>)	22.1	54.3	71.2
Spirulina	19.7	49.8	67.9
Sludge	16.3	40.1	58.3

SAFETY & HAZARDS (DRAWIO)

- ⚠ High Temperature (300–400 °C)
- ⚠ High Pressure (200 bar)
- ⚠ Flammable Gas (H₂, CO)
- ⚠ Corrosive Environment
- ⚠ Personal Protection Required

Risk Mitigation

- ✓ Pressure Relief Systems
- ✓ Gas Leak Detection
- ✓ Proper Ventilation
- ✓ PPE & Safety Training
- ✓ Regular Maintenance

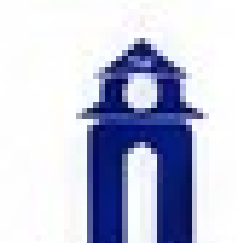
09 Acknowledgment and Limitations

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08 REFERENCES (Selected)

- Kumar, P. et al., Renewable & Sustainable Energy Review, 2021, 145, 111056.
- Duan, P. et al., Bioresource Technology, 2020, 123, 123546.
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SCAN ME



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